1) Intelligent Agents in a Smart Home Automation System

Introduction

An intelligent agent is a system that observes its environment through sensors and takes actions to achieve specific objectives. In a smart home, sensors collect information about motion, temperature, lighting and time. The agent uses this information to control lights, air conditioning and security system.

The four main types of intelligent agent are simple reflex, model-based, goal-based, and utility-based agents.

Simple Reflex Agent

A simple reflex agent selects action based on the current situation using predefined condition-action rules. It does not consider past experience or maintain information about previous state.

Working in a smart home:

- If motion is detected, turn on the lights.
- If the temperature exceeds 30°C, turn on the fan.
- If an unauthorized person is detected, activate the alarm.
- If no motion is detected for a specified time, turn off the lights.

Advantages:

- Simple to design and implement.
- Responds quickly to changes.
- Require less memory and computational power.

Disadvantages:

- Cannot remember past events
- cannot learn user preferences
- may make unsuitable decision when the environment is complex.

2) Model - Based Agent

A model - based agent maintains an internal model of the environment. It uses current sensor of the environment to understand the present situation.

working in a smart home.

- Remembers which rooms are occupied
- Tracks whether the user is at home or away.
- Uses previous temperature reading to understand environmental changes.
- consider door and window status before activity security system.

Advantages:

- works effectively in partially observable environment.
- maintain information about previous state.
- makes better decisions than a simple rule agent in changing situation.

2) Solving and unknown maze using UCS and IDS

A robot placed in an unknown maze must discover a route from its starting position to the exit without knowing the maze layout in advance

The robot can use uninformed search strategies, which explore possible paths without using additional information about how close a location is to the exit.

Two important uninformed search strategies are uniform cost search (UCS) and iterative Deeping Search

Uniform cost Search (UCS)

Uniform cost search expand the node with the lowest total path cost from the starting position. It uses a priority queue to select which node to explore next

Working of UCS

1) Place the starting position in a priority queue with a cost of zero.

2) Remove the position with the lowest accumulated cost

3) Check whether the position is the exit.

4) Calculate the cost of reaching each neighbor and add or update it in the priority queue.

Advantages of UCS

- Find the least-cost path when all step costs are non-negative
- Is complete when the relevant search conditions are satisfied
- Handles situations in which different movements have different costs.

Disadvantages of UCS

- ~~Rep~~ Repeats earlier searches at every new depth limit
- may perform unnecessary work when the exit is deep in the maze.
- Does not guarantee the least-cost path when movement costs differ.